ResoniCS — RER B — Planning du 2 au 6 novembre 2026

Les 22 tirets de l'énoncé sont repris mot pour mot et dans leur ordre. Les barres donnent la journée prévue et le nom en gras indique le responsable. Les deux cellules `Do not modify this cell` restent inchangées ; les compléments sont placés dans les fonctions et cellules autorisées.

Cas RER B. État de l'art : Mathis et Olympia (SOTA.md). Première plage proposée : roulement, 250–2 000 Hz. Traction/auxiliaires, 100–500 Hz, et crissement, 1 500–4 000 Hz, sont des cas complémentaires à valider avec la source, le matériau et le maillage.

Cadrage initial. To give the name to the startup, to define the objectives, the applications to develop, and the main strategy and organization. In addition, to define

Repère	Tâche de l'énoncé	Responsable / contributions	Lundi 2 nov.	Mardi 3 nov.	Mercredi 4 nov.	Jeudi 5 nov.	Vendredi 6 nov.
J1.1	the state of the art ;	**Olympia** Mathis ; synthèse disponible					
J1.2	the model to consider,	**Yann** Olympia, Joachim					
J1.3	the plage of frequencies of the interest, dimensions, noise sources ;	**Mathis** Yann, Olympia					
J1.4	formulate the properties of the chosen model, what is known about its solution ;	**Joachim** —					
J1.5	formulate optimization problem ;	**Antoine** Joachim					
J1.6	Theoretical study : considering that one boundary is open-air (the wave go through without coming back) derive the energy over the form and compare the result with the analogous result with a reflexive boundary[1].	**Joachim** Yann					
J1.7	to choose porous material(s) and its quantity ;	**Mathis** Antoine					
J1.8	find numerically the corresponding complex coefficient(s) $\alpha(\omega)$ for the porous matherial(s) ; to compare the performances if several ;	**Mathis** Yann					
J1.9	find energy derivative over the distribution of the porous inclusions ;	**Joachim** Yann					
J1.10	formulate/consider the numerical gradient descent method.	**Antoine** —					
J2.1	Study the energy on the chosen plage of frequencies for a fully absorbent wall of the plane shape and to define the local maxima ;	**Olympia** Yann, Mathis					
J2.2	Study the same question for different shapes of fully absorbent walls ;	**Olympia** Yann, Mathis					
J2.3	Compare the energy of the initial distribution of the porous material (this time the wall has not only absorbing parts but also reflective ones) with the energy for the fully absorbing wall.	**Olympia** Yann, Antoine					
J2.4	Optimize numerically for one or several of the most problematic frequencies (at which there is a maximum of the energy) ;	**Antoine** Yann, Olympia, Joachim					
J3.1	Analyze the obtained results to compare to the efficient performances on the fixed range of frequencies ; to compare to a fully absorbing wall ;	**Olympia** Yann					
J3.2	to optimize the energy at once at several different frequencies ;	**Antoine** Yann, Olympia, Joachim					
J3.3	to study which quantity of the porous material gives the most optimal result ;	**Mathis** Yann, Olympia, Antoine					
J4.1	To continue the analysis of the performances ;	**Olympia** Yann, Mathis, Joachim, Antoine					
J4.2	To conclude which product to produce/develop.	**Olympia** Yann, Mathis, Joachim, Antoine					
J5.1	To finalize everything	**Yann** Mathis, Olympia, Joachim, Antoine					
J5.2	Preparation of the defense	**Yann** Mathis, Olympia, Joachim, Antoine					
J5.3	Finalization of the rapport by the team.	**Yann** Mathis, Olympia, Joachim, Antoine					

FIGURE 1 – Planning des tâches de l'énoncé

[1] To model the open-air type of boundaries, it is convient to consider the Sommerfeld radiation condition of the form $\frac{\partial v}{\partial n} - ik \operatorname{Tr} v = 0$ with the notation $u(x,t) = e^{-ikt}v(x)$ for the propagative wave.

Validation quotidienne. Each day from 14h30 to 17h, the team/start-up presents the current state of the work (main results of the day, difficulties, strategy, ..) and validates its work. Each team member must approve and validate their portion of the work.

Vendredi après-midi. Defense by team the afternoon (slides 20 min, each person speaks, 10 min questions)